DOCUSCOPE™
GLOBAL

Version 1.8.4

User Guide

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English

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DocuScope Global, Version 1.8.4

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Welcome to DocuScope

DocuScope Global is a computer-aided text analysis tool that allows researchers to conduct both quantitative and qualitative analyses of how the designed reader experience is created by writers through the selection of micro linguistic composing patterns, i.e., words and phrases. Micro linguistic choices reference a heterogeneous but seamless array of reader cues seeded in texts spanning semantics, sentiment, and discourse register.

We have been curating these composing patterns for over two decades from thousands of texts across dozens of genres. Currently, we have collected over 70 million patterns of English into a dictionary that classifies these patterns into a hierarchy of nearly 50 top-level categories, 5,000 middle-level categories, and 60,000 low-level categories. Our curation efforts are conducted from the reader's point of view—i.e., we archive and classify reader-interpretable composing patterns from 1 to 11 consecutive words. Syntax and part of speech are essential for accurate classification in our dictionary and we rely on information from both syntactic position and part of speech categorization to make our classifications. For example, "the butcher" (butcher = noun) is classified as a character type. "She butchered him" (butcher = verb) is classified as negative aggression.

The English Default Dictionaries within DocuScope were developed to fill in a meso-layer of language that had been missing in standard treatments of rhetoric and written expression. The framework of a meso-layer of language in the rhetorical arts was first introduced by David Kaufer and Brian Butler in *Rhetoric and the Arts of Design* (1996) and refined further to capture textual-mediated experience in *Designing Interactive Worlds with Words: Principles of Writing as Representational Composition* (Kaufer and Butler 2000). According to Kaufer and Butler in the latter book, standard writing pedagogies acknowledge the grammatical relations (words, phrases, clauses) that hold a sentence together, the coherence relations across sentences that hold a paragraph together, and the situational constraints that define language as "appropriate" to its context and often "genre appropriate" when the features of language contributing to appropriateness have been rigorously observed and codified for students as a genre. Kaufer and Butler (2000) argued for the importance of a missing middle layer between grammar and genre that connects internal sentence parts to the recurring experiences that help constitute the designed "reader experience" of language. This middle layer constitutes the intricate patterning through which words in combination produce a range of reader experiences required for precisely controlling the language.

This intricate patterning of words leading to semantic dynamics and stable rhetorical classification is the focus of the book The Power of Words: Unveiling the Speaker and Writer's Hidden Craft, published in 2003 by David Kaufer, Suguru Ishizaki, Jeff Collins, and Brian Butler. That book exposes the hidden craft by which single words can undergo radical shifts across meaning, register, sentiment, and genre, and still be rhetorically classified, predicted, and controlled as a contributor to the designed reader experience. For example, consider the word "bargain" in English. While we think of "bargain" as an intact word, the meaning, sentiment, and register it conveys depends on the various word sequences in which it participates and the sequential position, grammatical form, and grammatical function of the word within that sequence. Readers may come across the phrase "insane bargain"

and know they are likely reading hyped-up commercial advertising. They may come across "bargaining sessions" and "collective bargaining" and know they are reading about social conventions. They will recognize the phrase "opening the bargaining with" as an initial act of negotiation. The phrase "no bargain" will be read as a putdown, and "more than anyone bargained for" an indication of an unexpected difficulty encountered. Depending on the position, form, and function of the word in the stream of discourse, "bargain" contributes to larger language patterns in drastically different ways. For example, in DocuScope's Default Dictionary, "Driving a hard bargain" is classified with "tough negotiator" and "haggling." "Holding up one's end of the bargain" is classified with "responsibility and dependability." In the ever-shifting environment in which writers make composing decisions, individual words are unruly but the patterns in which they circulate across the language are rhetorically relatively well-governed and can be systematically classified according to the larger rhetorical function they serve. It is this deeper level of organization where words in context are classified to predict reader experience that Kaufer and Butler associate with a meso-layer of language. This meso-layer, according to Kaufer and Butler, governs how individual words bond with other words to make meanings predictable and, as such, stable candidates for appropriate language use.

Three levels of dictionary organization within DocuScope's Default English Dictionary are summarized below.

Cluster	Clusters are the highest level of rhetorical categorization. Clusters act as a broader way to categorize rhetorical patterns. DocuScope organizes patterns first into clusters, which then branch into Dimensions
Dimensions	Dimensions are the second level of rhetorical categorization. Dimensions refine clusters based on certain rhetorical patterns.
Language Action Types (LATs)	Language Action Types are the smallest level of categorization. LATs are usually defined by individual words, or a smaller string of words than clusters and dimensions.

The following is a partial list of the many classification types in which "bargain" contributes to an English writer's meaning, sentiment, genre, and register in DocuScope's current default dictionary. If you start with the left side of the list you will see that bargain appears in categories designated at the highest level as Information ("bargain sale"), Metadiscourse ("into this bargain, we might further add…"), Negative ("wouldn't bargain"), Positive ("more than one could have bargained for"), Public [legal] Terms ("plea bargain"), and Strategic ("bargained for"). By tracking the streams and the

⋮
InformationTopics ▸ InformationTopicsCommercialAdcopy
InformationTopics ▸ InformationTopicsDimensionGeneral
InformationTopics ▸ InformationTopicsMoney
MetadiscourseCohesive ▸ MetadiscourseCohesiveAdditive
Negative ▸ NegativeActResistDenyDimensionGeneral
Negative ▸ NegativeActResistResistant
Negative ▸ NegativeEmotionDifficult
Negative ▸ NegativeEmotionDimensionGeneral
Positive ▸ PositiveValuesBargain
PublicTerms ▸ PublicLawPlea
PublicTerms ▸ PublicLawPlea
Strategic ▸ StrategicCompetitionBargain
Strategic ▸ StrategicCompetitionNegativeHaggle
⋮

nature of the streams in which even a single word flows in a text, a researcher gets a better sense of the larger rhetorical functions any single text or collection of texts is meant to serve. This sense of overall rhetorical function can become increasingly precise, we have hypothesized, as we can classify all the words of a text and collate the aggregate streams of the text and the rhetorical functions that emerge from that aggregation. Over two decades have been spent building the dictionary because we have sought to do for tens of thousands of words of English what we have done for "bargain."

With DocuScope Global's Default Dictionary, you can start analyzing your text or a corpus immediately. DocuScope Global supports three ways to analyze texts.

- Analyze a single text
- Analyze a group of text
- Analyze a set of groups

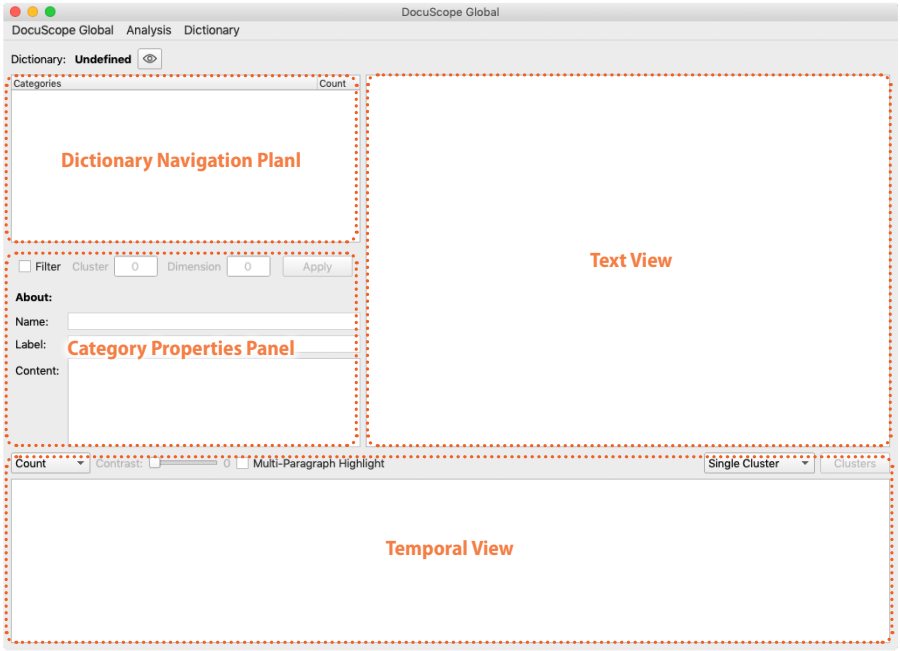
A single text can be analyzed using DocuScope Global's visualization tools. If you are interested in analyzing one or more groups of texts, you can export the results as a CSV file and analyze the output in a statistic application. We recommend you to start visualizing a single text to familiarize yourself to the tool as well as the dictionary's classification scheme.

DocuScope Global also allows researchers to build their own customized dictionaries according to their own language theories to use in the analysis. DocuScope dictionaries are written with three levels of hierarchy: Clusters, Dimension, and LATs (language action types). Clusters represent the top level of organization, dimensions are more refined in scope, and LATs are more specific and defined. LATs are where you write the patterns for your dictionary. You should first try building small dictionaries (which you can think of as coding schemes). DocuScope Global supports the rapid development of mutually exclusive dictionaries (dictionaries where every pattern is uniquely classified in one LAT and there is no overlap.

Overview of the User Interface

DocuScope Global's user interface consists of four major panels:

- Dictionary Navigation Panel
- Text View
- Category Properties Panel
- Temporal View



Load a Dictionary

To use DocuScope, you must first load a dictionary. You can use the default dictionary provided by us, or if desired, you can create your own dictionary. DocuScope's Dictionary Editor allows the user to systematically explore and examine domain-specific composing patterns and create a new custom dictionary. Refer to the Create a New Dictionary section for more details.

Load a Dictionary from a DAT File

You must load a dictionary before analyzing a text. The Default Dictionary is usually distributed with DocuScope as a .dat file.

1. Launch DocuScope Global.
2. Choose Dictionary > Load from DAT.
3. Select a dictionary file (.dat file) and click Open.

   Loading a dictionary usually takes about 2-3 minutes, but it may take longer on computers with limited memory size.

Use Load from Folder... to load your custom dictionary.

Analyze a Text

DocuScope Global supports Microsoft Word files (.docx) or plain text files (.txt, either UTF8 or ASCII files).

⚠ ***Before analyzing a text, you must load a dictionary into DocuScope Global.***

Open a Single Text

1. Choose Analysis > Open File...
2. Specify a file and click Open.

   The Text View shows the text, and the Dictionary Navigation panel shows the count for each of the clusters included in the dictionary.
3. To visualize composing patterns, click a category in the Dictionary Navigation panel. Click the arrow button next to the category name to select sub categories (i.e., dimensions and LATs).

   The Category Property panel displays a definition of the selected category.

Filter Language Categories

DocuScope Global allows you to filter language categories by their counts. By default, all the categories are shown, including categories with no matches (count = 0). You can set a minimum number of occurrences of language patterns to exclude categories with few results.

1. Select the Filter checkbox.
2. To filter at the cluster level, enter a number into the Cluster text field and click Apply. Clusters with instances fewer than the given number will not be displayed.

   To filter at the dimension level, enter a number into the Dimension text field and click Apply. Dimensions with instances fewer than the given number will not be displayed.

Analyze Patterns Over Time

The Temporal View panel allows you to examine how language patterns are used over time (from the beginning to the end of the text). The Temporal View displays how the selected cluster and corresponding dimensions are used in each paragraph in the text.

Graphing Options

1. Click the Graph Options drop down menu.
2. Select one of the graphing options.

The following table summarizes the graphing options.

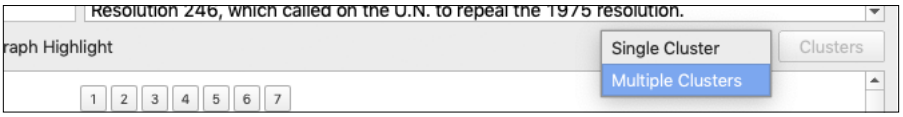
Option	Definition
Count	The total number of patterns per category in each paragraph. This option is the default.

Frequency	The total number of patterns per category divided by the total number of words in each paragraph.
Log Scale	The common logarithm of the count. Frequencies can be misleading when there are very short paragraphs. The log scale is used to compensate for this problem. Higher frequencies that are created by short sentences are proportionally suppressed using a log scale.
Binary	Presence of or absence of patterns in each paragraph.

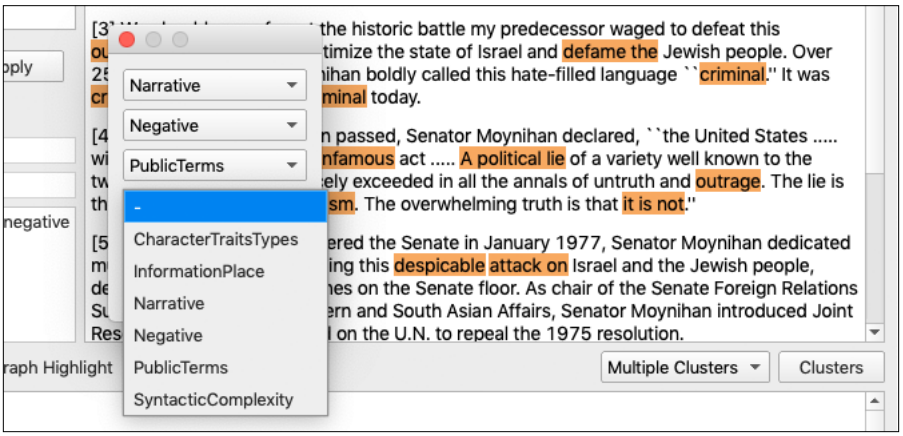
Display Multiple Clusters

By default, the Temporal View graph displays a single cluster along with its dimensions. Alternatively, you can display graphs for multiple clusters.

1. Select Multiple Clusters from the drop down menu in the Temporal View Panel.



2. Click the Clusters button, select the desired clusters from the Drop Down Menus, and press OK.



The drop down menus will display all the clusters shown in the Category dialog box. If you have an active filter, the excluded clusters will not be visible in this menu.

Highlight Paragraphs

The Temporal View allows you to highlight a paragraph in the Text View by clicking the paragraph buttons in the Temporal View.

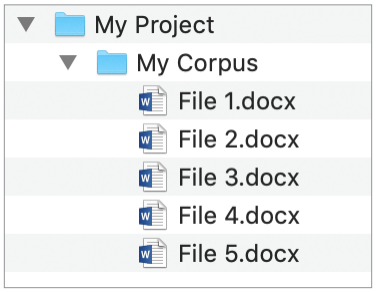
You can highlight multiple paragraphs by clicking the Multi-Paragraph Highlight checkbox.

Analyze a Corpus

DocuScope Global allows you to generate a spreadsheet (CSV file) that contains the DocuScope analysis of multiple files—i.e., a corpus. Your corpus may include a single group of files, or multiple groups of files.

Before analyzing your corpus, you must organize the files in a specific folder structure.

Set up a Corpus Folder for a Single Group of Texts



1. Create a new folder for your project.
2. Inside that project folder, create another folder for your corpus.
3. Copy all the text files to the corpus folder.

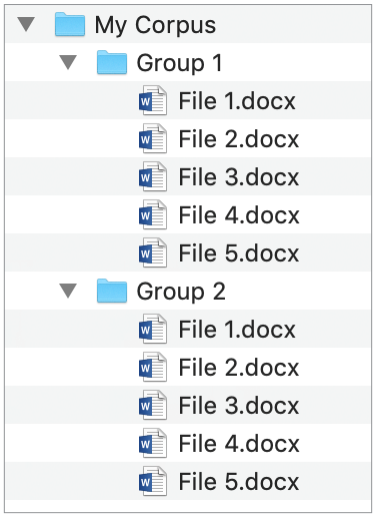
We recommend that the project folder is not in the application folder for DocuScope Global.

Generate a CSV file from a Single Group of Texts

1. Launch DocuScope Global and load a dictionary.
2. Select Analysis > Generate CSV from Corpus.
3. Specify a corpus folder and click Open.

An output folder will automatically be created in the folder that contains your corpus folder. A CSV file is generated for each level of the dictionary.

Set up a Corpus Folder for Multiple Groups of Texts



1. Create a new folder for your project.
2. Inside the project folder, create another folder for your corpus.
3. Inside the corpus folder, create folders for groups.
4. Copy all the text files to the group folders.

We recommend that the project folder is not in the application folder for DocuScope Global.

Generate a CSV File from Multiple Groups of Texts

1. Launch DocuScope Global and load a dictionary.
2. Select Analysis > Generate CSV from Corpus.
3. Specify a corpus folder and click Open.

An output folder will automatically be created in the folder that contains your corpus folder. A CSV file is generated for each level of the dictionary.

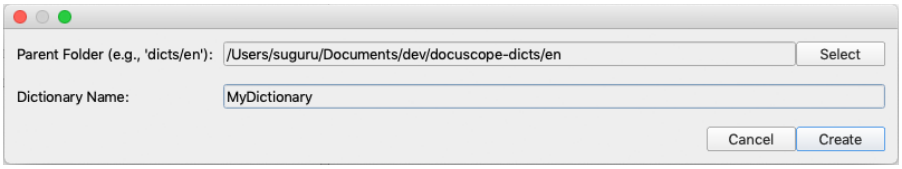
Create a New Dictionary

DocuScope allows analysts to explore and create a new domain-specific dictionary to use with a text.

Create A Dictionary

1. Select Dictionary > New...

   This will open a dialog for creating a new dictionary.



2. Select a folder where the new dictionary will be saved.
3. Enter a dictionary name, and click Create.

   A new dictionary sub-folder is created. This folder contains all the files that are needed for the dictionary.

⚠ ***No files in the dictionary folder should be edited manually. Refer to the Edit a Dictionary section to learn how to make changes to a dictionary.***

Adding Categories

When a new dictionary is created, a cluster is automatically added along with a dimension and an LAT. Default names will be given to all three categories.

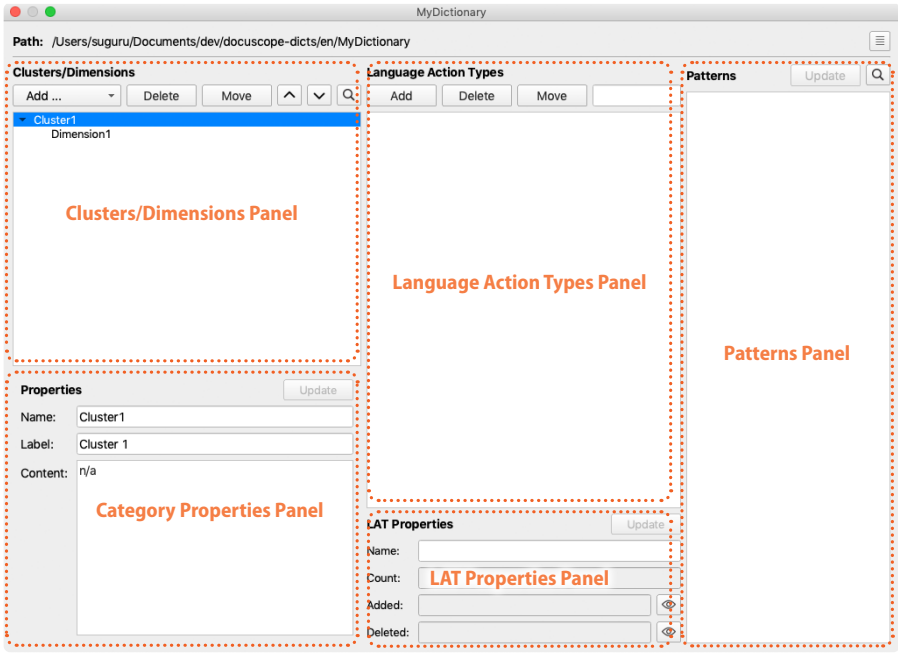
The following properties need to be entered for each of the clusters and dimensions in your dictionary.

Property	Definition
Name	A unique name is required for each category. No dimensions and clusters can have the same name. The name should not include spaces and should not start with a number. Conventionally, names are written in CamelCase.
Label	The label is a user friendly name for the category.
Content	A brief description must be entered for each category.

Overview of the Editor

DocuScope Global's user interface consists of five major panels:

- Category Properties Panel
- Language Action Types Panel
- Language Action Type (LAT) Properties Panel
- Patterns Panel



Edit a Dictionary

Open a Dictionary for Editing

1. Choose Dictionary > Open in Editor.
2. Select the dictionary folder and click Open.

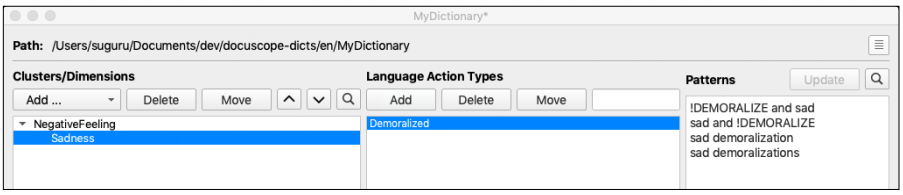
⚠ ***DAT files cannot be opened in the editor. Dictionaries are protected intellectual property, and the base language pattern files must be available to edit a dictionary.***

Add Clusters and Dimensions

1. In the Clusters/Dimensions Panel, click Add... and select Add Cluster or Add Dimension.
2. In the Category Properties panel, enter a name for the cluster (or dimension) in the Name field and click Update.

Add an LAT to an Existing Dimension

1. In the Clusters/Dimensions panel, select a cluster, then a dimension to which a new LAT will be added.



2. In the Language Action Types panel, click Add.

   A new untitled LAT will be added.

3. Add a name in the LAT Properties panel below and click Update.

Edit Cluster or Dimension Properties

1. Select a cluster or dimension in the Clusters/Dimensions panel.
2. In the Category Property panel, change the name, label, and enter a definition of the cluster / dimension.
3. Click Update.

During text analysis, these properties are valuable for users who are not familiar with your dictionary scheme.

Move a Dimension to Another Cluster

1. Select a dimension in the Cluster/Dimensions panel.
2. In the pop up menu, select the destination cluster and click OK.

Move an LAT to Another Dimension

1. Select a language action type in the LAT panel.
2. In the pop up menu, select the destination dimension and click OK.

Add/Remove Language Patterns to an LAT

1. Choose the cluster, dimension, and LAT.
2. In the LAT Patterns panel, add or remove patterns.
3. Click Update.

Create/Edit Word Classes

Dictionary authors often need to enter patterns with words in multiple forms. For example, consider the following patterns with the word verb "demoralize" in four different forms.

- demoralize the nation
- demoralized the nation
- demoralizes the nation
- demoralizing the nation

Writing these patterns can be tedious, and they may take up a lot of space in the LAT Pattern Editor. To simplify the entry of these patterns, DocuScope Global allows you to define a special entry called "word class." For example, you may define a word class called "DEMORALIZE" for the different forms of the verb "demoralize." Then, the following single line,

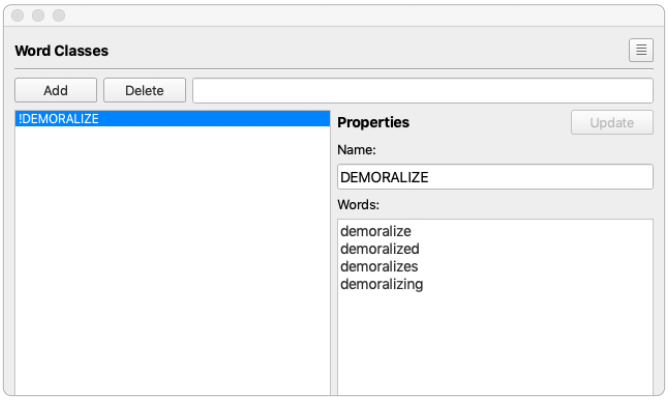
!DEMORALIZE the nation

will be expanded into four unique patterns when your dictionary is loaded into DocuScope Global.

Notice that the exclamation mark (!) is used in front of the word class. You must use the exclamation mark to indicate a word class.

⚠ ***Word classes cannot contain spaces, and cannot start with a number. Word classes are conventionally written in all upper case letters.***

1. Click the Menu Icon and select Edit Word Classes.
2. Edit or delete existing word classes, or create new word classes.



⚠ ***In the name field, you must not enter the exclamation mark, which should be used when you enter patterns in the Dictionary Editor.***

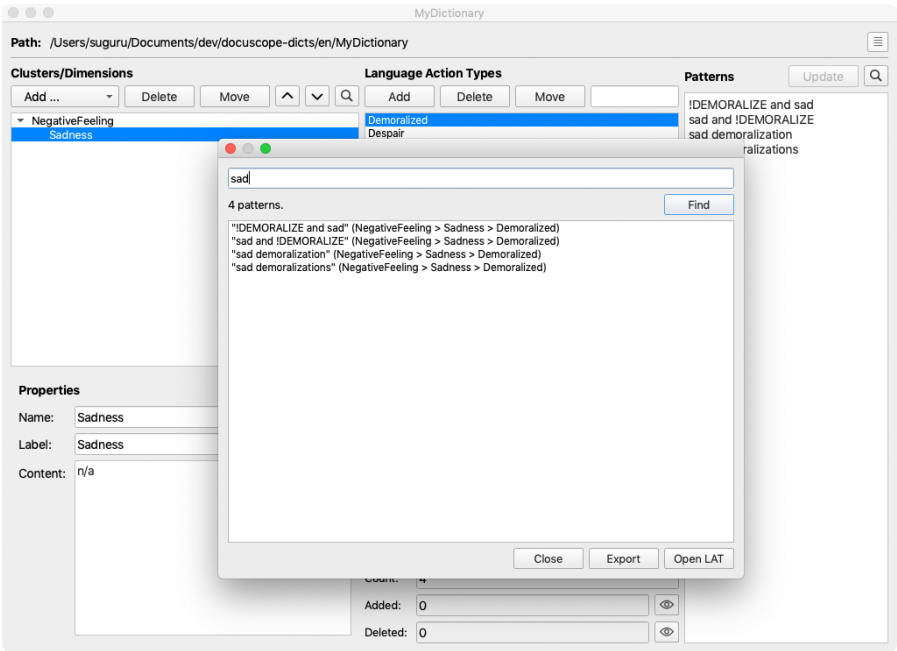
3. Once you are done with your edits, close the dialog box.

Finding a Pattern in LATs

If you are interested in finding an LAT that contains a specific pattern, you can use this feature.

⚠ ***You must load the entire dictionary by selecting Load Dictionary from the menu to enable this feature. Loading a dictionary may take a few minutes if your dictionary is large.***

1. Click the Menu icon, and select Find in LATs…
2. Enter a pattern you are looking for, and click Find.
3. In the result panel, select the LAT and the pattern you are interested in, and press the Open LAT button.



4. Press the Export button to export the search results in a CSV file.

Verify a Dictionary

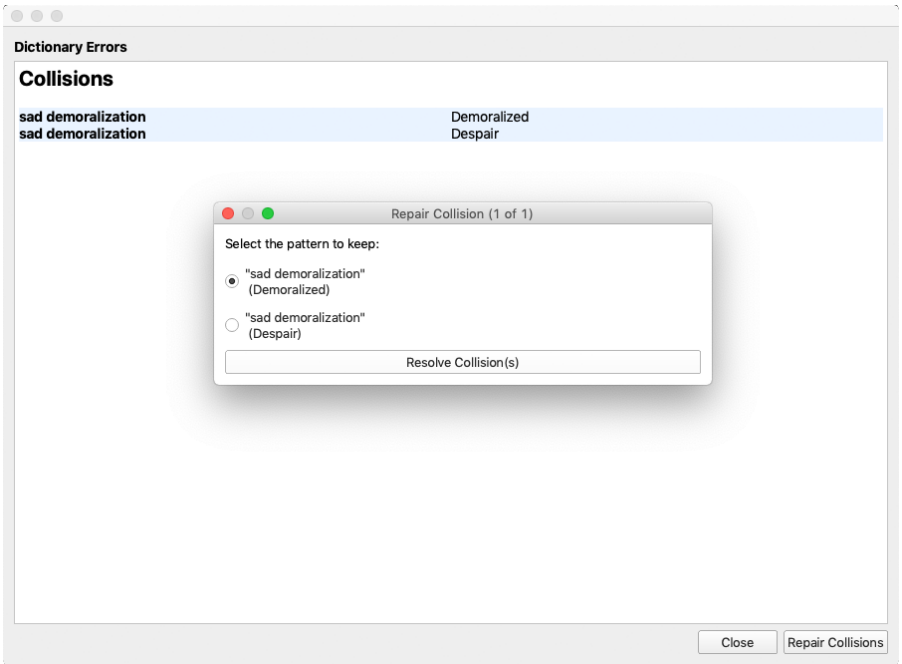
DocuScope Global allows you to verify the changes you have made to the dictionary. As the dictionary becomes larger, you might accidentally enter a pattern in multiple categories. This is called a "collision", which is a common problem. The verifying tool helps dictionary authors to fix these errors.

1. Click the Menu Icon and click Verify.

   The following items are presented in a separate dialog box:

   - A list of single word collisions.
   - A list of multi-word collisions.
   - Undefined word classes.
   - Duplicate word classes.

2. If collisions are detected, click Repair to start the automatic tool that resolves single and multi-word collisions.



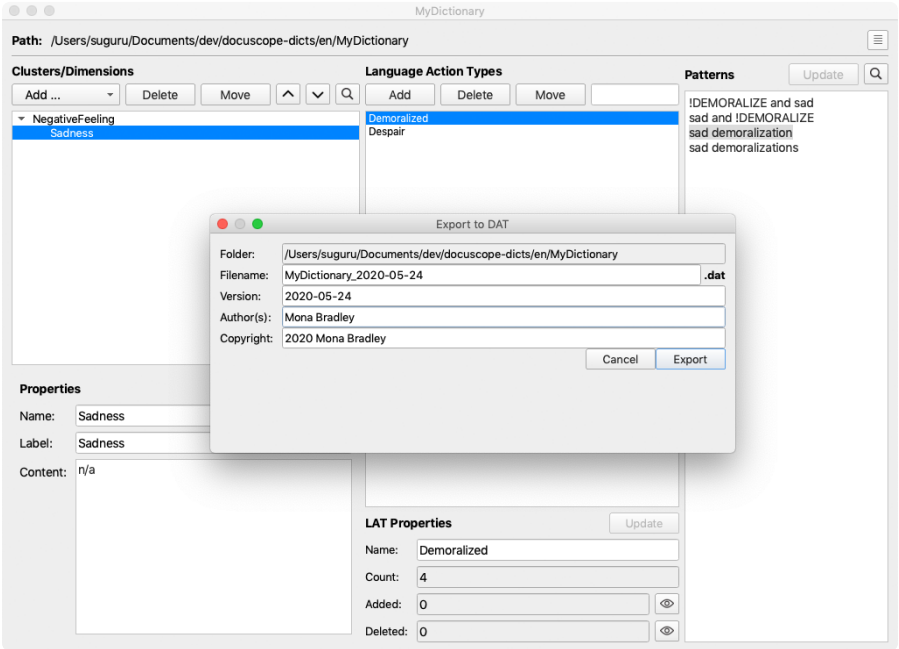
⚠ ***Undefined and duplicate word classes must be fixed manually.***

⚠ ***Depending on the size of the dictionary, the verification process may take a long time. If your dictionary contains over several thousand patterns, the verification may take several hours.***

Export a Dictionary

The .dat file is a single file that contains all the dictionary pattern files.

1. Click the Menu icon and select Export to DAT.



2. Fill out the information about the dictionary in the dialog box.
3. Click Create.

   A new .dat file is created in the parent folder of your dictionary folder.

⚠ ***Depending on the size of the dictionary, it may take a few seconds to a few minutes. If your dictionary contains over several thousand patterns, it may take 30 minutes or more.***

Iterative Dictionary Development

Language patterns included in a dictionary are often iteratively "harvested" from a corpus of texts. The following is a typical dictionary development process supported by DocuScope Global.

1. Create an initial dictionary (i.e., draft) in the Dictionary Editor.
2. Edit the dictionary (i.e., add/revise categories, add/remove patterns).
3. Save the dictionary.
4. Close the Dictionary Editor.

   You will be prompted to load the dictionary into the analysis environment. Click OK.
5. Open a text file to test the dictionary.

   Highlight patterns in the text by selecting categories in the Category Navigation panel. Find patterns that are not included in the dictionary and list them.
6. From the Dictionary menu, select Edit <your dictionary name>. This will open the dictionary in the Dictionary Editor.

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